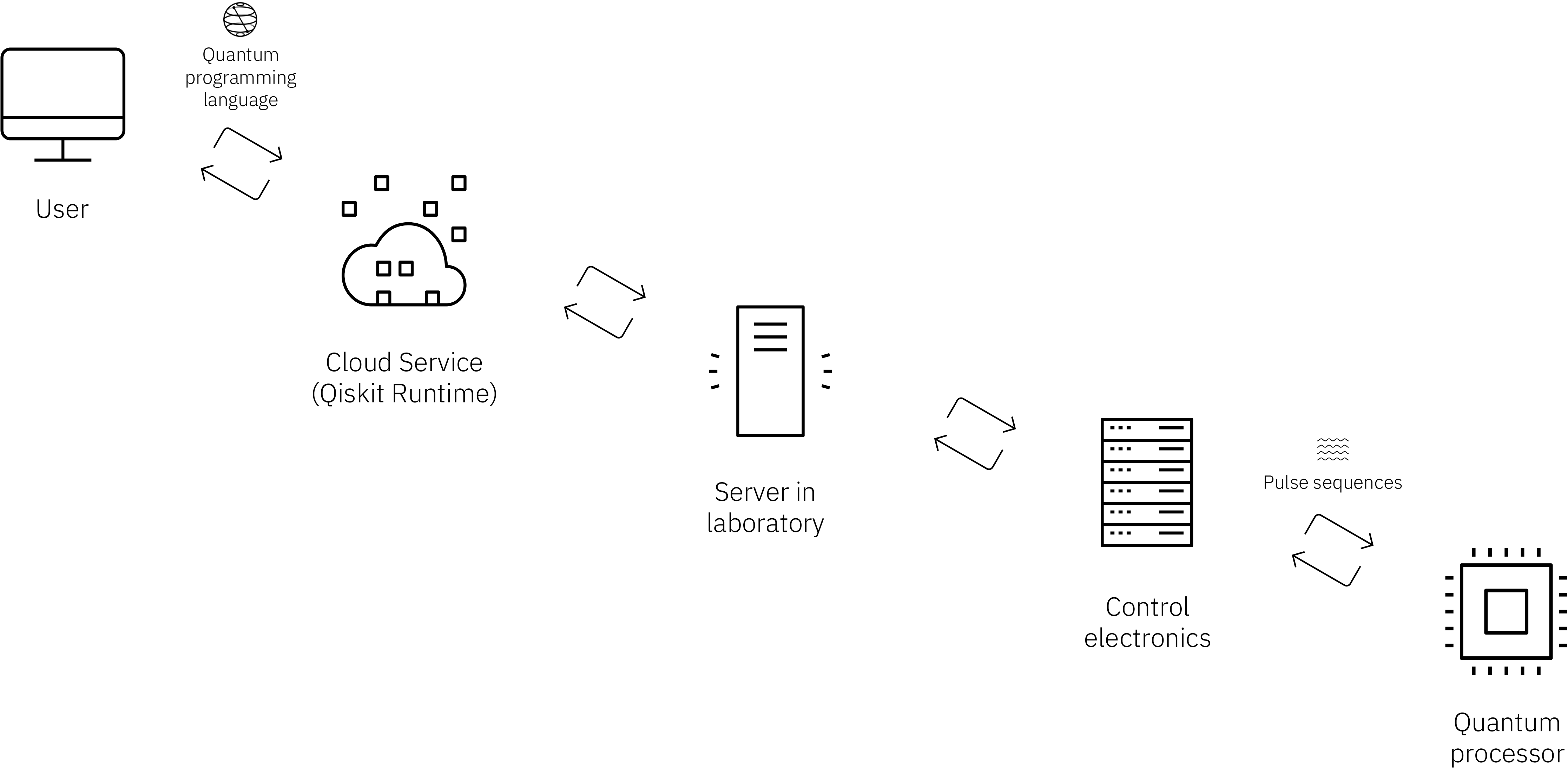


Introduction to Qiskit

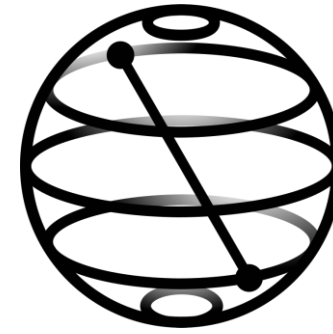
Somi Park

Client Engagement Manager,
IBM Quantum

Anatomy of a quantum computing service



What is Qiskit?



Qiskit is an open-source SDK for working with quantum computers at the level of extended quantum circuits, operators, and primitives.

<https://github.com/Qiskit/qiskit>

<https://quantum.cloud.ibm.com/docs/en/guides>

Qiskit Runtime (IBM Quantum Compute Service) is a cloud-based service for executing quantum computations on IBM Quantum hardware.

<https://github.com/Qiskit/qiskit-ibm-runtime>

The **Qiskit Ecosystem** is a collection of software and tutorials that build on or extend Qiskit.

<https://www.ibm.com/quantum/ecosystem>

Workflow – Qiskit Patterns

Map

Starting with a classical problem and figuring out how to map it to a quantum computer

Optimize

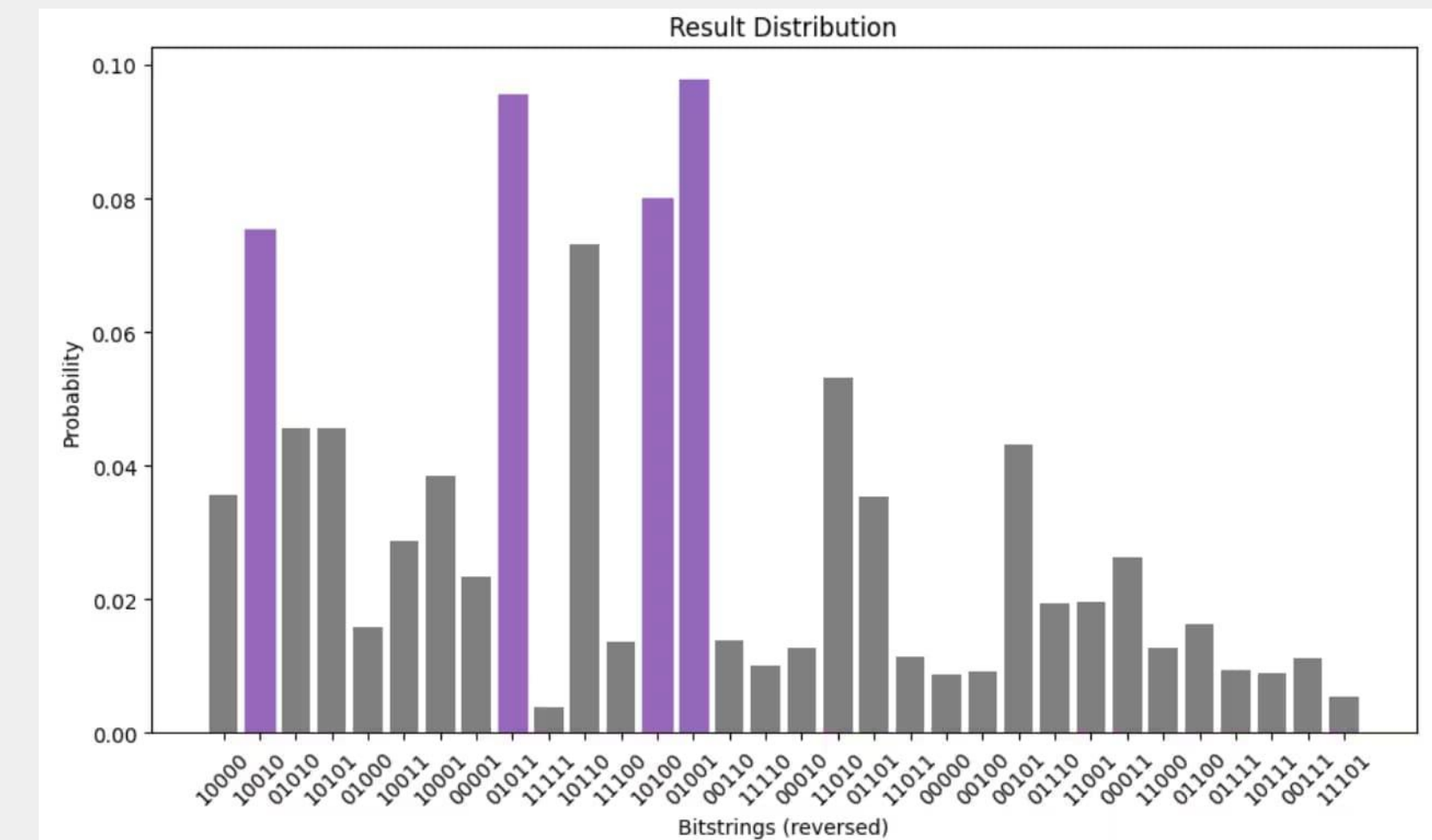
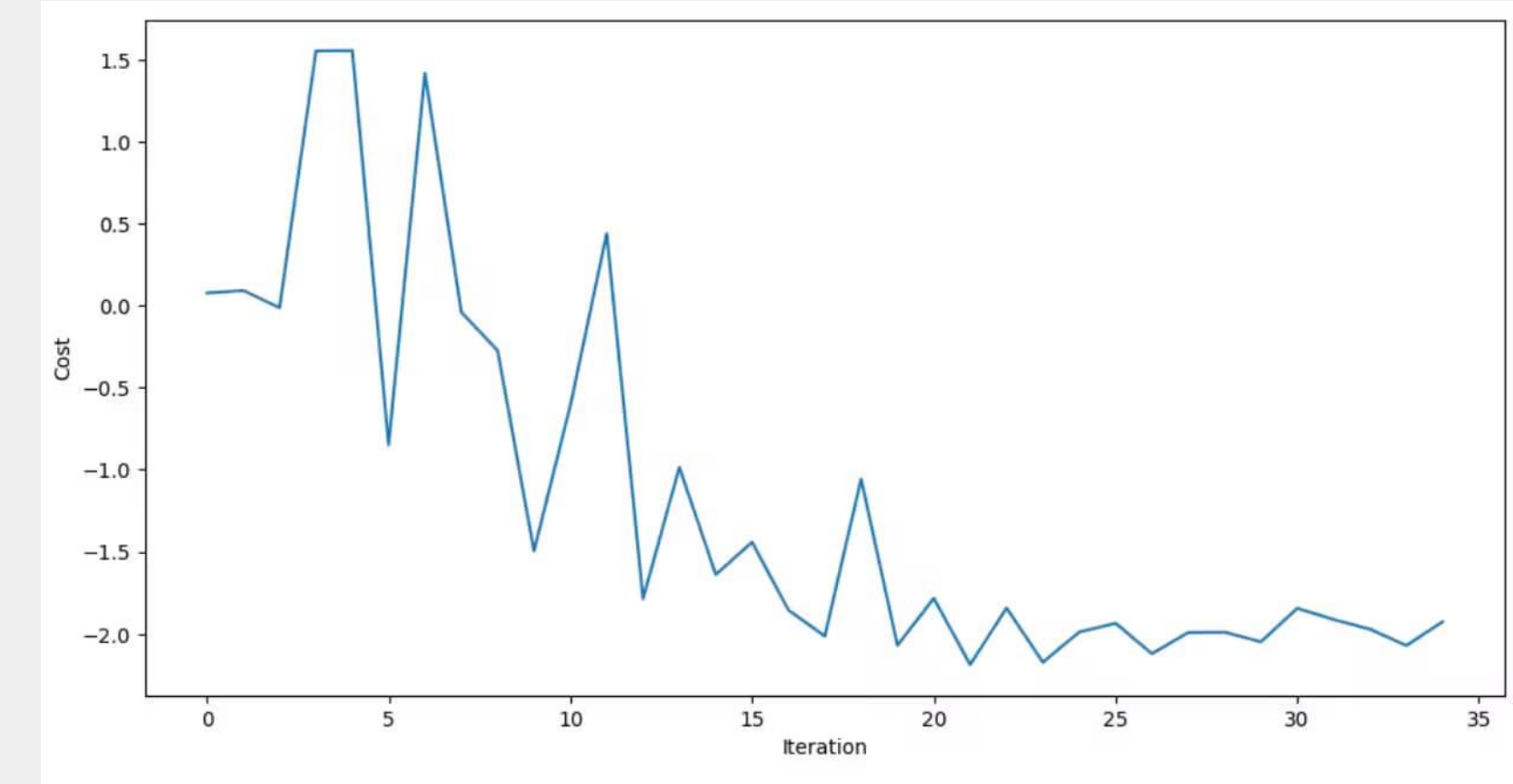
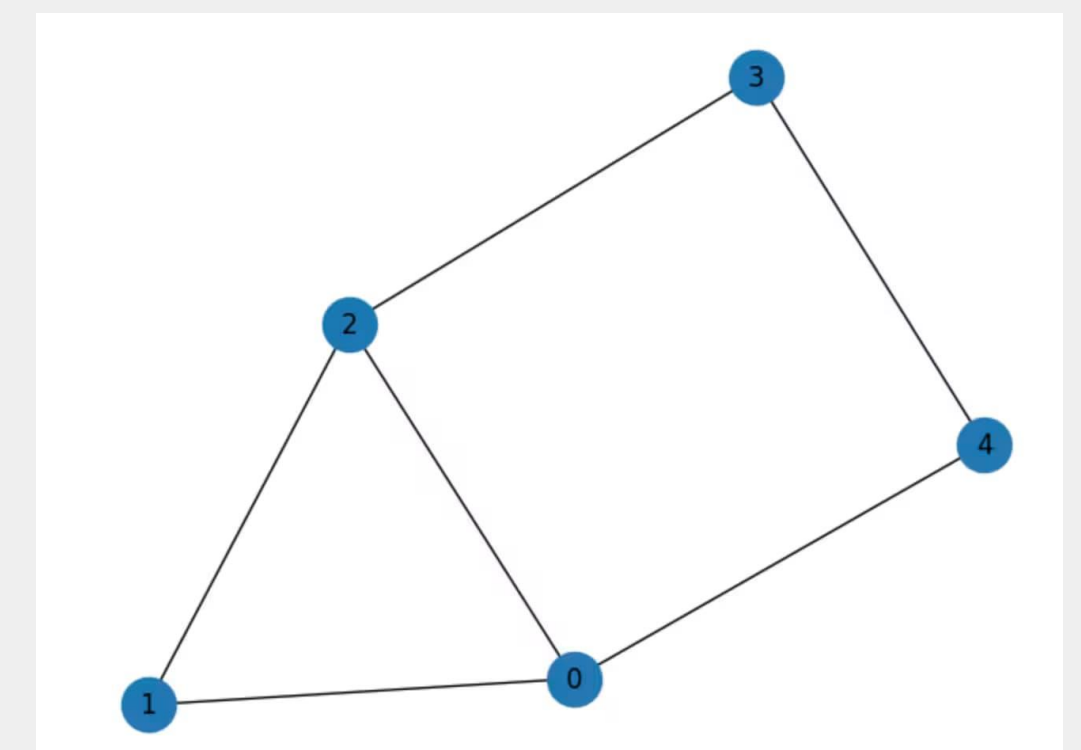
Use Qiskit's transpiler to route and layout the circuit to physical qubit hardware, convert to basis gates of the hardware, and reduce the number of operations

Execute

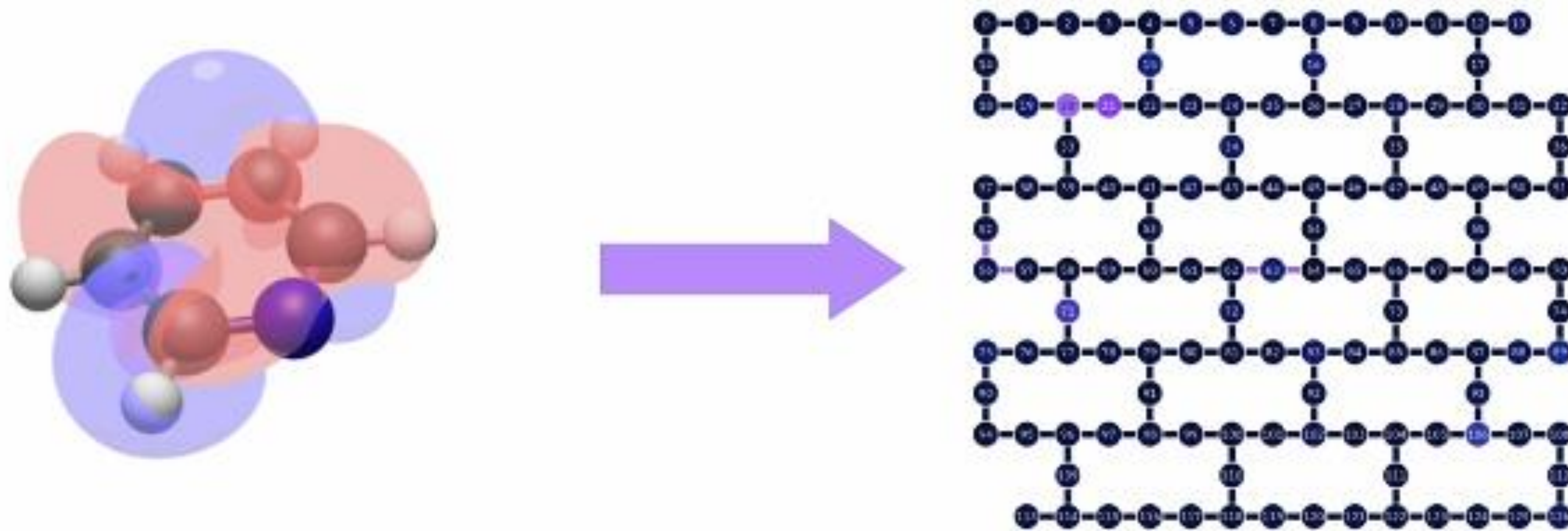
Use the Qiskit Runtime primitives to run your quantum circuits on quantum hardware

Post-process

Evaluate your circuits and data



1. Mapping real problems to circuits

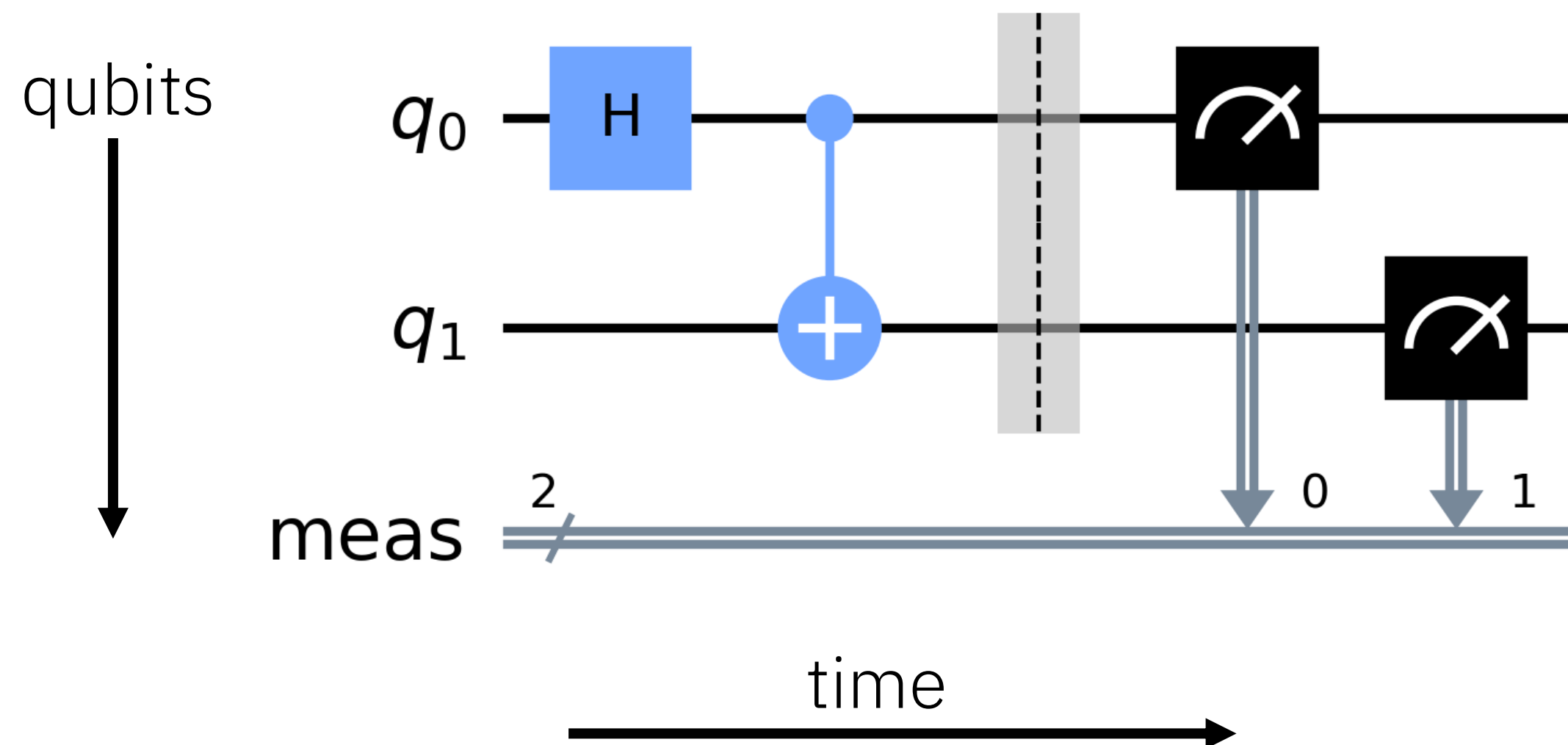


To map a problem:

- Start with numerical representation
- Map to quantum computing representation
- Choose quantum algorithm

Quantum Circuits

The foundation of quantum programs are quantum circuits, which consist of operations - including gates, measurement - that manipulate qubits in the quantum computer.



[IBM Quantum Composer](#)

```
from qiskit import QuantumCircuit, QuantumRegister

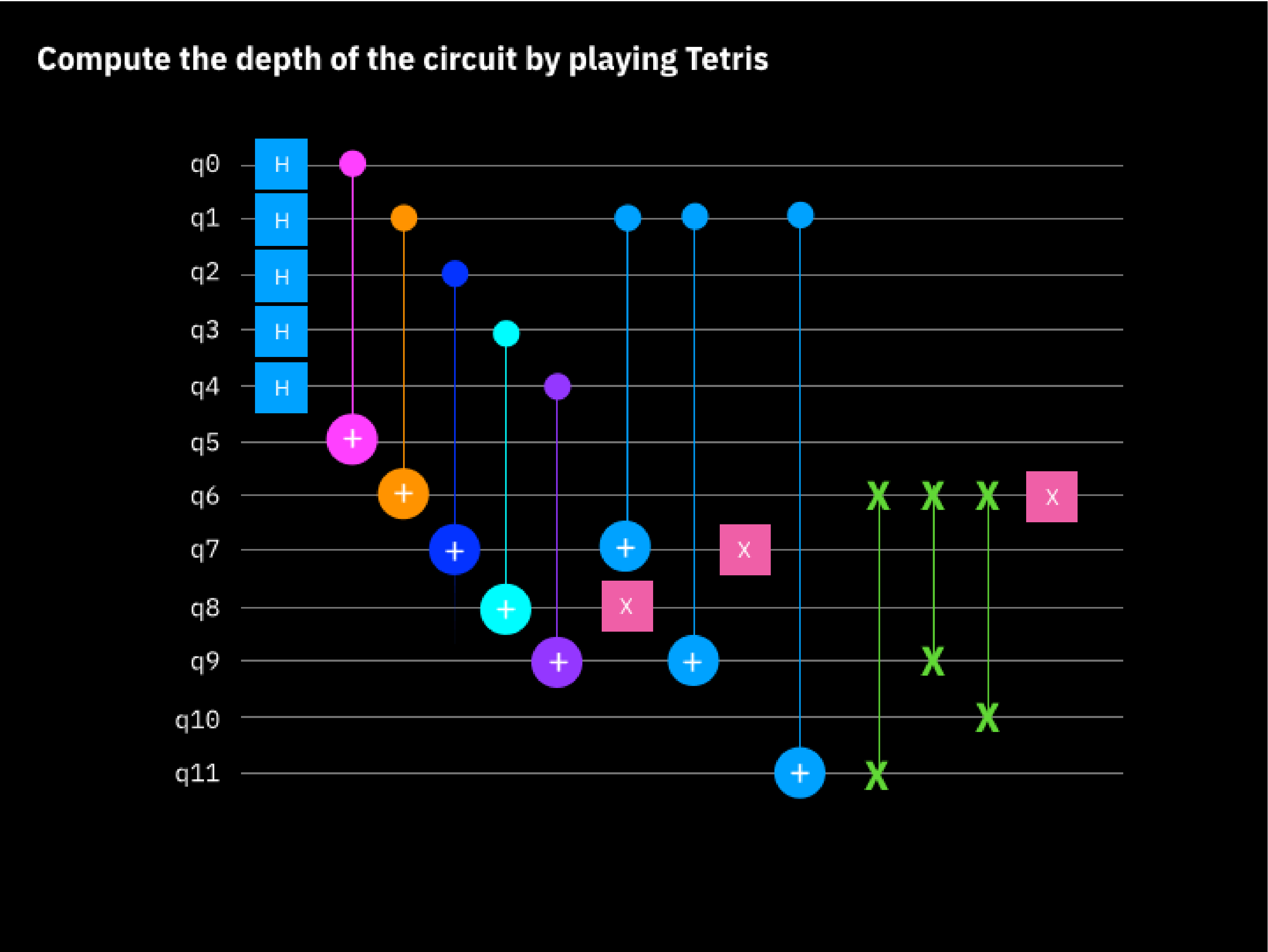
qubits = QuantumRegister(2, name="q")
circuit = QuantumCircuit(qubits)

q0, q1 = qubits
circuit.h(q0)
circuit.cx(q0, q1)
circuit.measure_all()

circuit.draw("mpl")
```

- Initialize a register of qubits
- Add the qubits to a circuit
- Perform operations on those qubits

Circuit Depth



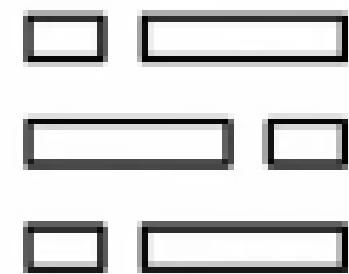
2. Optimize for hardware

Transpile your abstract circuit into a circuit that can run on target hardware

Real
quantum
devices are
subject to
constraints

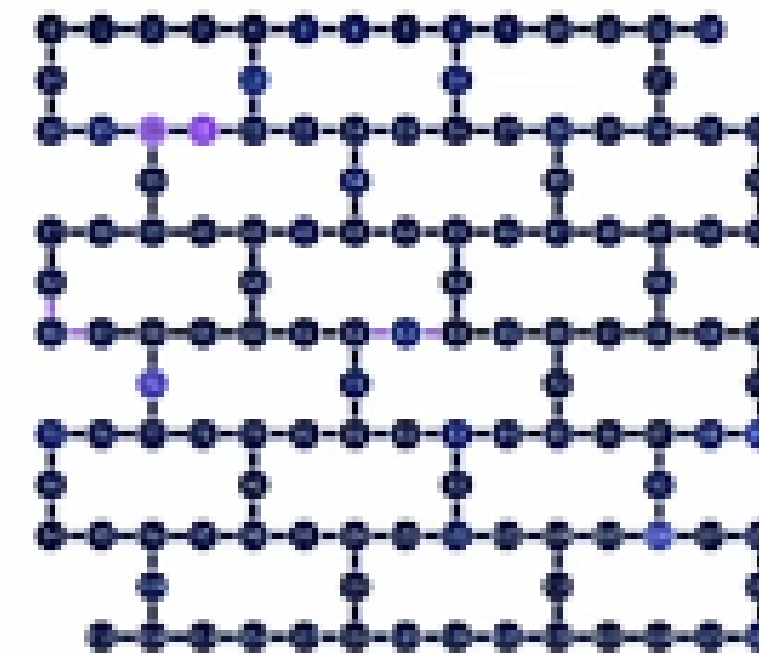
Basis gate set

Only a limited set of gates can be executed directly on the hardware. Other gates must be rewritten in terms of these basis gates.



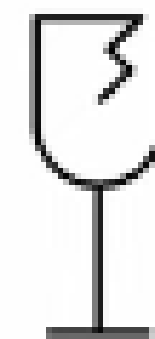
Qubit connectivity

Only certain pairs of qubits can be directly interacted with each other.



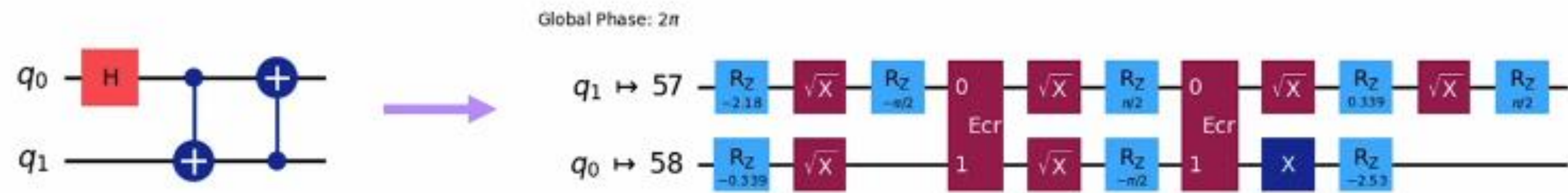
Errors

Each operation has a chance of error, so circuit optimizations can greatly affect performance.



2. Optimize for hardware

Transpile your abstract circuit into an ISA (instruction set architecture) circuit

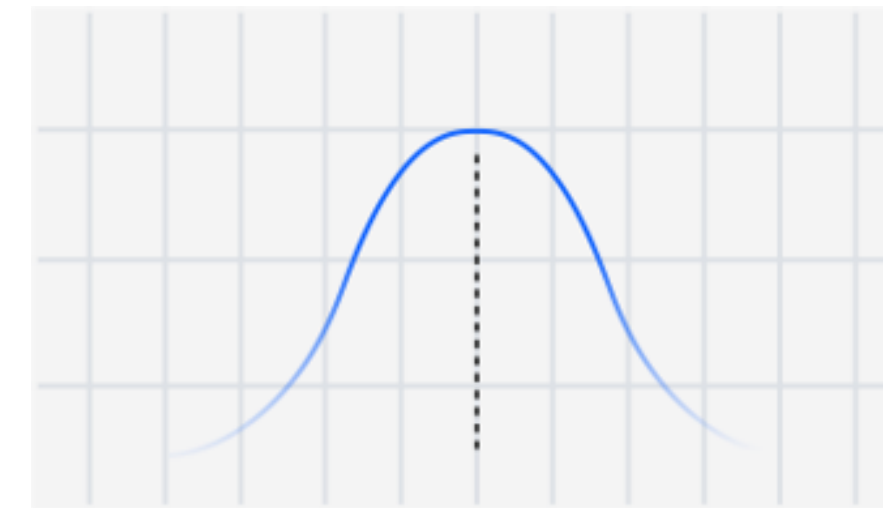


Term	Definition	Orchestra analogy
pass	a standalone circuit or metadata transformation	an instrument
pass manager	a list of transpiler passes grouped into a logical unit	an instrument section
staged pass manager	a list of pass managers, with each one representing a discrete stage of a transpilation pipeline	the conductor

3. Execute

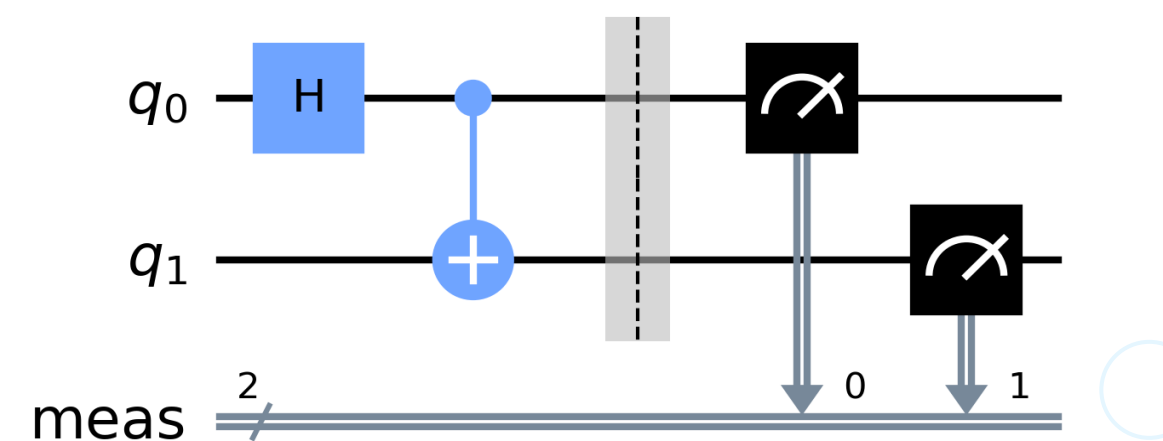
Primitives encapsulate the output of a quantum circuit

Sampler primitive

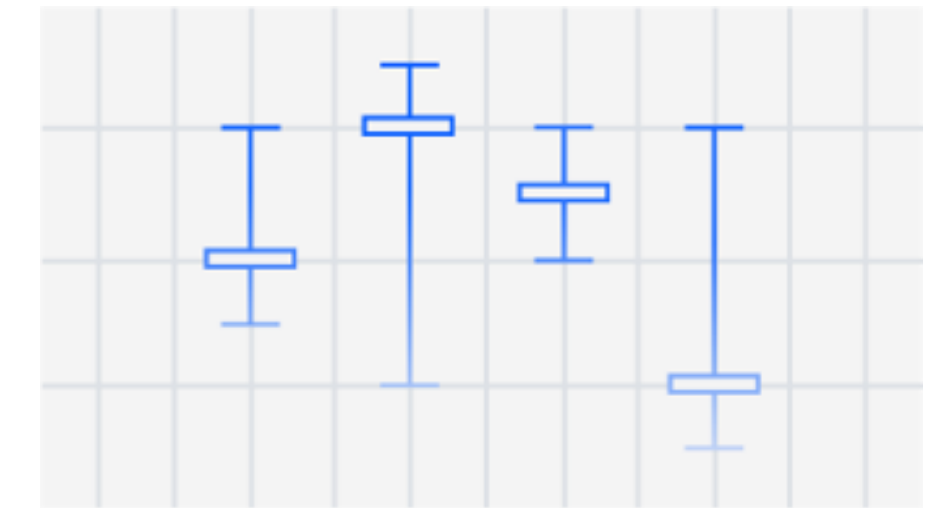


Output is mapping of bitstrings to counts, e.g, $\{ '0':12, '1':9 \}$

Circuit should include measurements

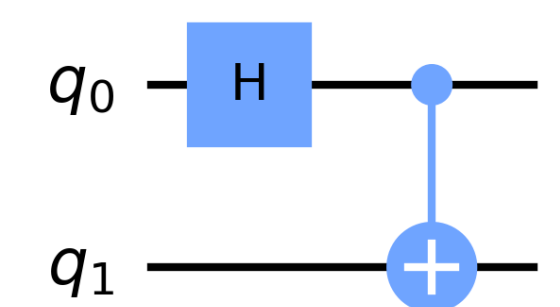


Estimator primitive



Output is the expectation value of an observable (e.g. cost function of an optimization problem)

Circuit should not include measurements



3. Execute

You can
customize
Qiskit Runtime
behavior

Shots

The number of
measurements.

More shots reduce
statistical error but
increase running time.

Error suppression

Use dynamical decoupling
to reduce decoherence
during execution.

Caveat: requires delays
between gate executions,
so it's not always possible.

Error mitigation

Reduce the effects of
device noise after
execution.

- Twirled Readout Error
eXtinction (TREX)
measurement twirling
- Zero Noise Extrapolation
(ZNE)

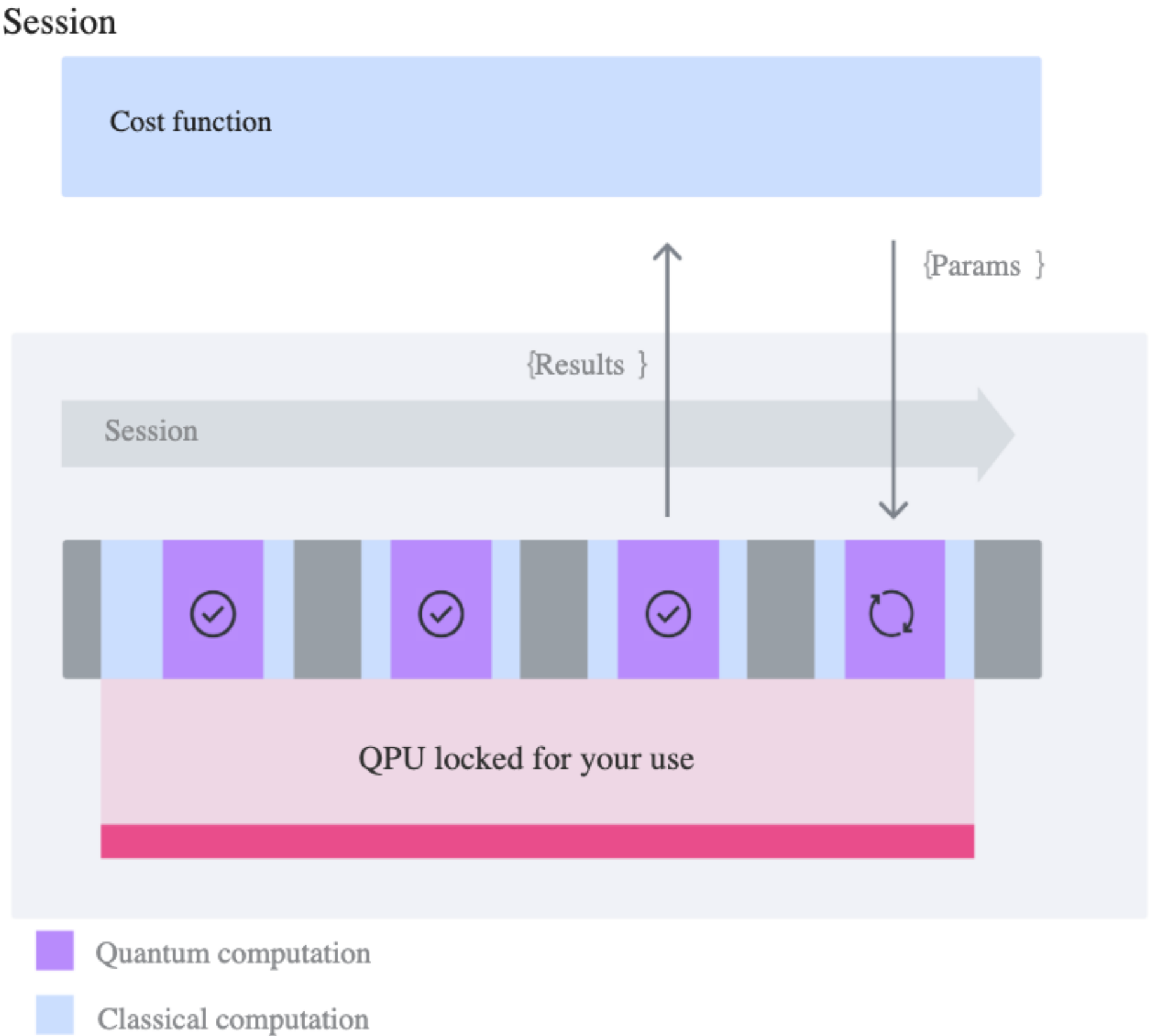
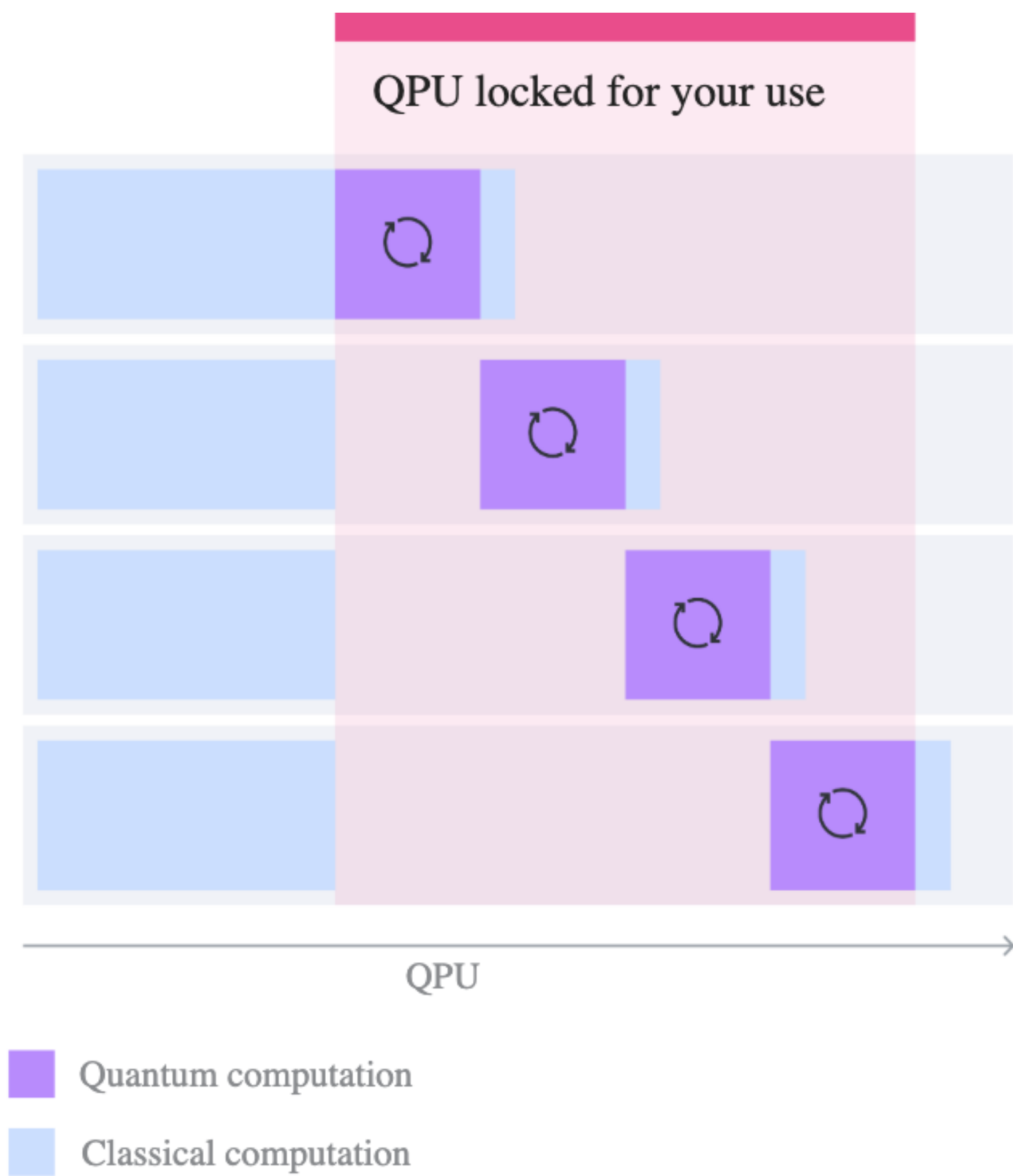
Downside: computational
overhead

3. Execute – Execution modes

Job mode
- Single job

Batch mode
- Multiple concurrent jobs

Session mode
- Iterative workload



4. Post-process

Error Mitigation refers to techniques that allow users to reduce circuit errors by modeling the device noise at the time of execution

- Twirled readout error extinction (TREX)
- Zero noise extrapolation
- Probabilistic error cancellation

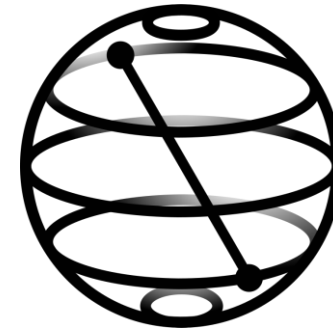
https://quantum.cloud.ibm.com/docs/en/guides/error-mitigation-and-suppression-techniques#enable-trex-by-using-the-measure_mitigation-option

~~["1111", "1010", "1010", "1110"]~~



["1010", "1010", "1110"]

Getting started with Qiskit



```
pip install 'qiskit[visualization]'  
pip install qiskit-ibm-runtime
```

To access IBM hardware, set up your credentials through one of:

- IBM Quantum Platform
- IBM Cloud

<https://quantum.cloud.ibm.com/docs/en/guides/quick-start>

IBM Quantum Documentation

<https://docs.quantum-computing.ibm.com/>

Access the documentation for Qiskit and IBM Quantum services.

IBM Quantum Learning

<https://learning.quantum-computing.ibm.com/>

Learn the basics of quantum computing, and how to use IBM Quantum services and systems to solve real-world problems.

Qiskit YouTube

<https://www.youtube.com/qiskit>

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